

From Lung to Brain:

Linking Chest CT Tumour Characteristics with Brain MRI Metastatic Patterns in Adenocarcinoma, Squamous Cell, Small Cell and Neuroendocrine Lung Cancer

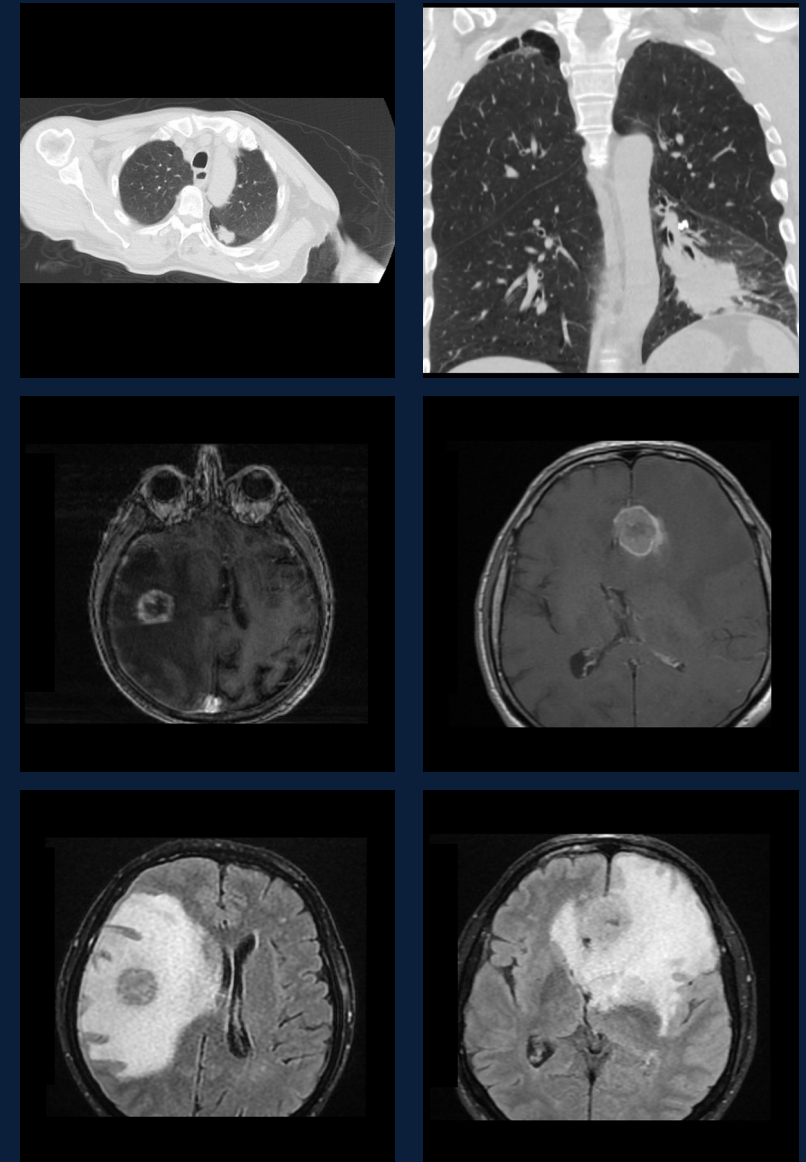
[First Name Surname]¹, [Co-author]¹, [Co-author]²

¹ [Department of Radiology, University / Hospital, Istanbul, Türkiye] ² [Department of Thoracic Oncology / Neurosurgery]

Corresponding author: [email]

Keywords: brain metastases · lung cancer · adenocarcinoma · squamous cell · SCLC · MRI · CT

Ethics committee approval and written patient consent for scientific publication obtained.



Chest CT · T1+C · FLAIR (SCLC, left; adenocarcinoma, right)

Purpose & Objectives



To illustrate how brain MRI features of metastases vary with the histological subtype of the primary lung cancer.

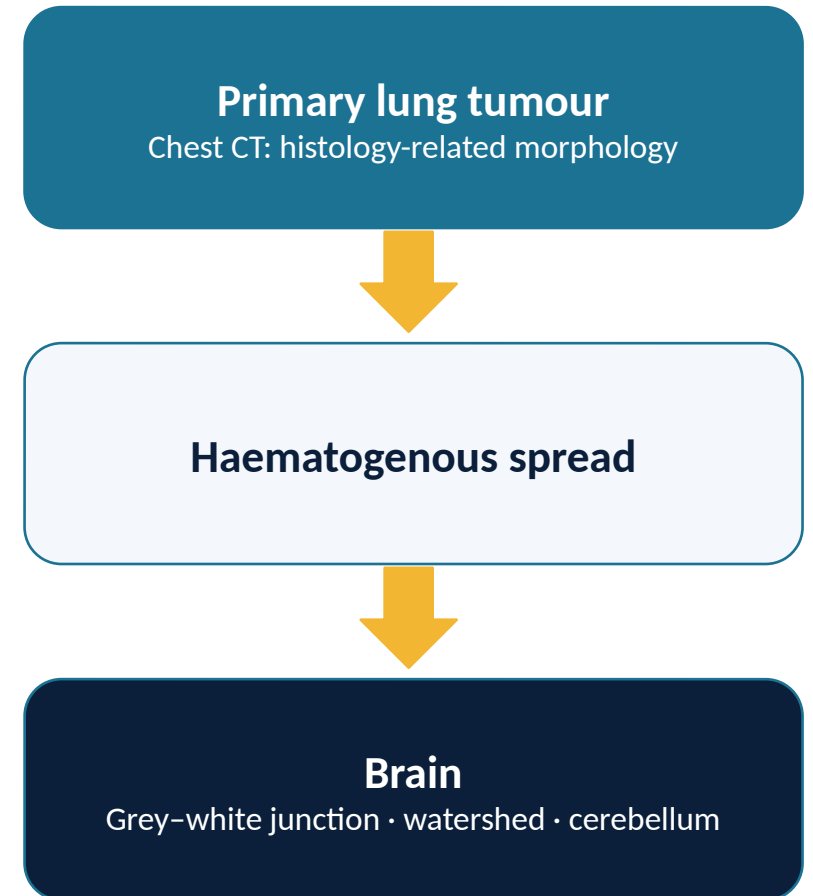
- 1 Recognise the chest CT appearance of adenocarcinoma, squamous cell, small cell and large cell neuroendocrine carcinoma.
- 2 Describe multi-sequence MRI patterns (T1+C, T2/FLAIR, DWI) of the associated brain metastases.
- 3 Compare number, location, enhancement, oedema and diffusion behaviour between subtypes.
- 4 Identify mimics and pitfalls, e.g. an incidental meningioma alongside metastases.
- 5 Understand how imaging supports staging, PCI/SRS decisions and follow-up.

Introduction & Clinical Background

- Lung cancer is the leading source of brain metastases (BM), about 40–50% of all BM [1,2].
- Cumulative BM risk is higher in small cell lung cancer (SCLC) than in non-small cell lung cancer (NSCLC) [1,2].

Histology shapes risk and behaviour

- **Adenocarcinoma:** commonest NSCLC subtype with BM; risk increased with EGFR/ALK alterations [8].
- **SCLC:** highly proliferative; frequent BM; MRI surveillance and prophylactic cranial irradiation (PCI) are debated [5,6].
- **Squamous cell carcinoma:** BM less frequent [10]; primaries often central, cavitating or necrotic.
- **LCNEC:** large cell neuroendocrine carcinoma, biologically closer to SCLC [4].
- Contrast-enhanced MRI is the most sensitive modality; lesion number, size and site decide surgery, stereotactic radiosurgery (SRS) or whole-brain radiotherapy (WBRT) [6,7].



Materials & Methods



Patient selection

n = 10, histologically proven lung cancer with brain metastases

Chest CT review

Primary tumour: site, size, margin, cavitation

Brain MRI review

Number, site, enhancement, oedema, diffusion

Pattern comparison

Descriptive, by histological subtype

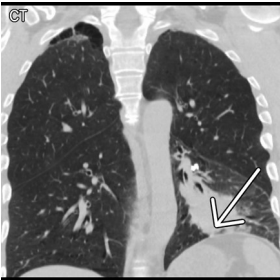
- **Design:** retrospective, descriptive pictorial case series.
- **Approvals:** ethics committee; written informed consent for publication of anonymised images.
- **Cohort:** adenocarcinoma (ADC) n = 5; small cell carcinoma (SCLC) n = 3; LCNEC n = 1; squamous cell carcinoma (SCC) n = 1.
- **Illustrated:** six representative cases (ADC-2, ADC-3, SCLC-1, SCLC-2, LCNEC-1, SCC-1).
- **Inclusion:** histological diagnosis, chest CT and brain MRI available.
- **Imaging:** chest CT (axial, coronal; lung and mediastinal windows) and brain MRI: axial post-contrast T1 (T1+C), T2, FLAIR, DWI with apparent diffusion coefficient maps.
- **Analysis:** consensus review; descriptive statistics only.

Primary Lung Tumour Profiles

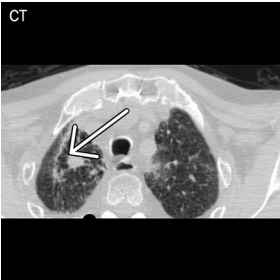
Case	Histology	Primary tumour (chest CT)	Brain metastasis (MRI)
ADC-2	Adenocarcinoma	Left perihilar mass, lower lobe collapse (coronal CT)	Left medial frontal
ADC-3	Adenocarcinoma	Small spiculated right apical nodule; emphysema	Right temporal
SCLC-1	Small cell carcinoma	Lobulated left upper lobe mass, posterior, pleura-based	Right paramedian frontoparietal
SCLC-2	Small cell carcinoma	Small left upper lobe nodule, posterior, pleura-based	Right frontoparietal
LCNEC-1	Large cell neuroendocrine	Left apical-posterior soft-tissue mass abutting pleura	Left frontal
SCC-1	Squamous cell carcinoma	Very large central mediastinal mass; tracheal compression; bullous emphysema	Left cerebellar hemisphere

Not illustrated: adenocarcinoma $n = 3$, small cell carcinoma $n = 1$. Sides follow radiological convention.

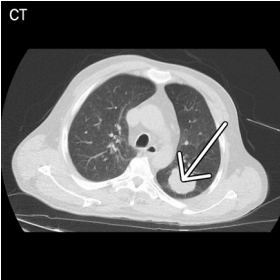
Primary tumours (white arrows)



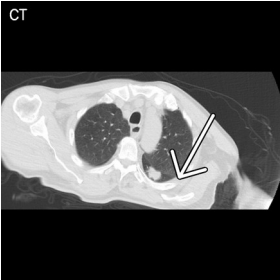
ADC-2



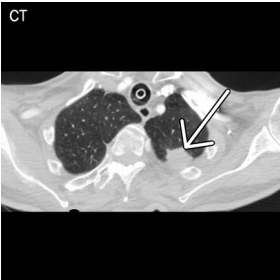
ADC-3



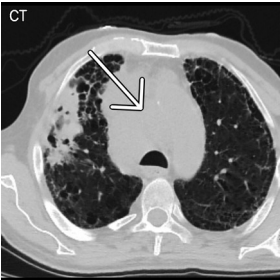
SCLC-1



SCLC-2

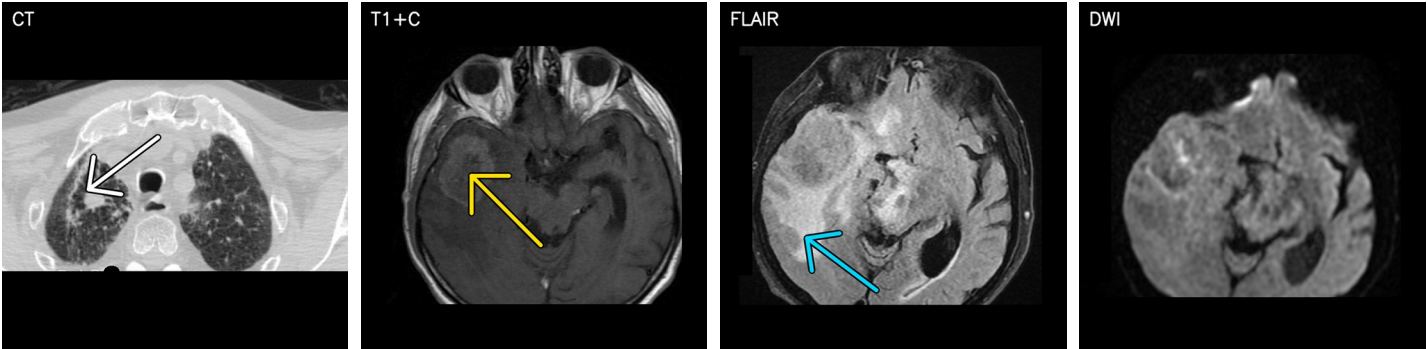


LCNEC-1

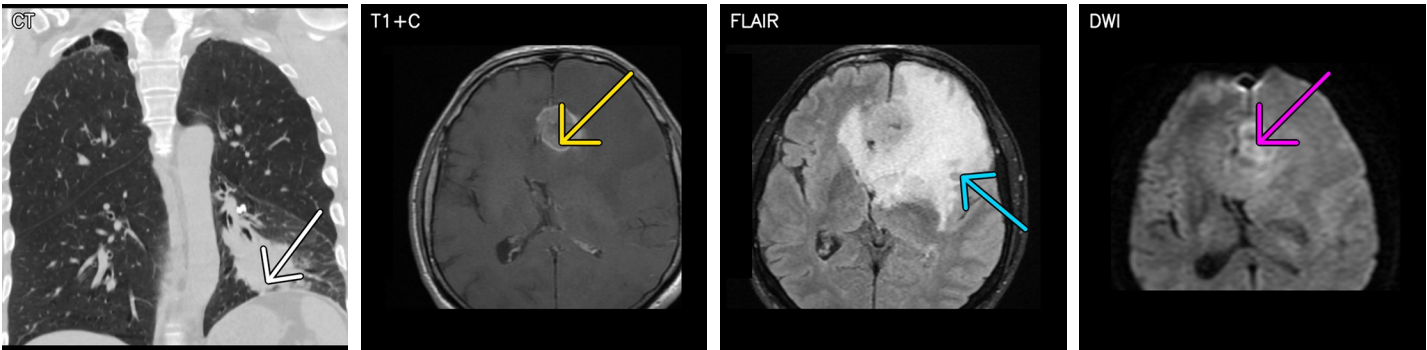


SCC-1

Neuroimaging Spectrum: Adenocarcinoma Brain Metastases



ADC-3: chest CT, axial T1+C, FLAIR, DWI



ADC-2: chest CT (coronal), axial T1+C, FLAIR, DWI

Key imaging features

Feature	ADC-3	ADC-2
Site	Right temporal	Left medial frontal
T1+C	Ill-defined ring/heterogeneous	Thick irregular ring, necrotic centre
FLAIR	Extensive vasogenic oedema	Extensive vasogenic oedema
DWI	No restriction	Rim restriction

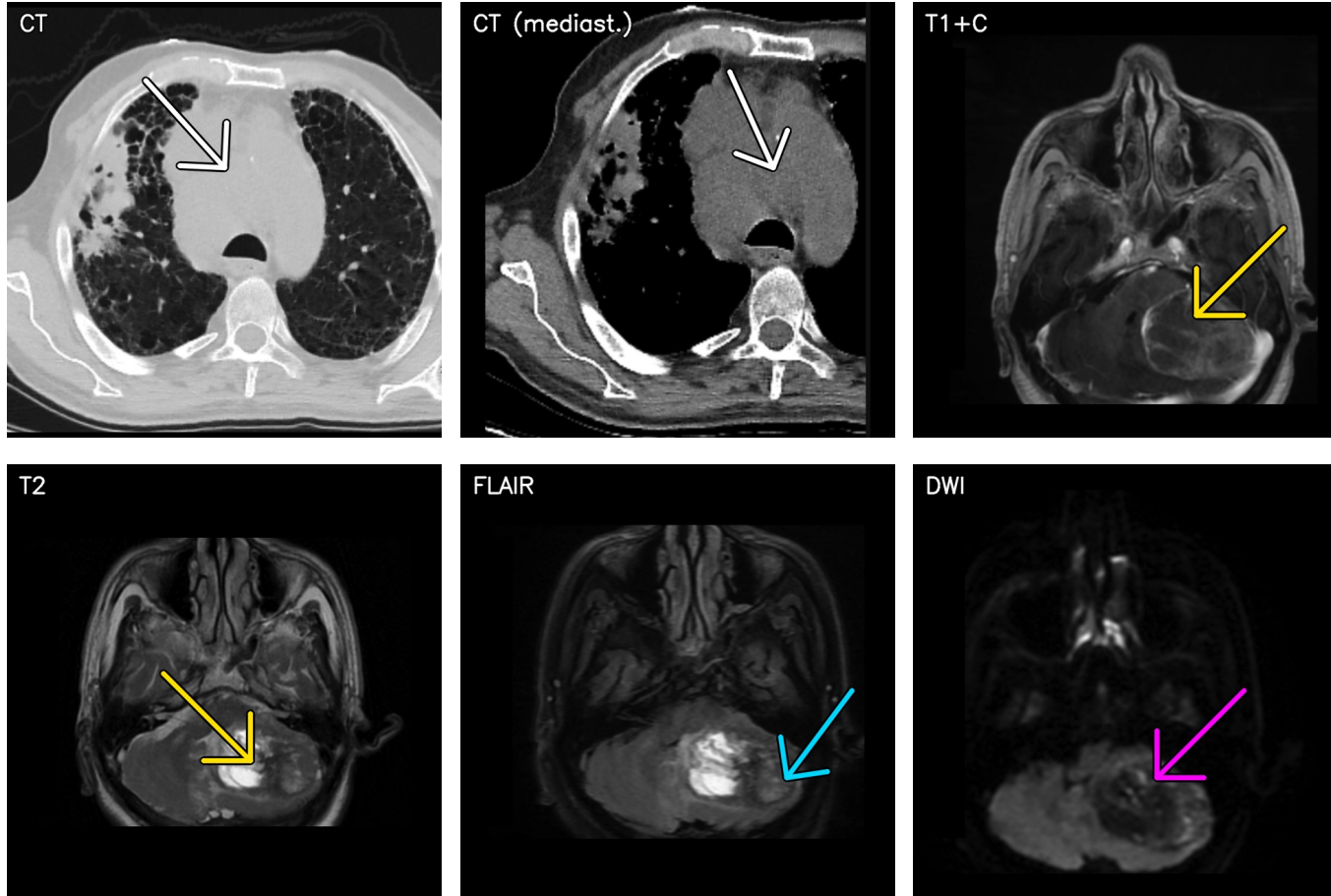
Teaching points

- Small primaries can present with large brain lesions; oedema is disproportionate to lesion size.
- Correlate with EGFR/ALK status where available [8].

Arrows

- Enhancing lesion
- DWI restriction
- Vasogenic oedema
- Primary lung tumour

Neuroimaging Spectrum: Squamous Cell Carcinoma Brain Metastases



SCC-1: chest CT (lung, mediastinal window), T1+C, T2, FLAIR, DWI

Key imaging features

- **Primary:** very large central mediastinal mass compressing the trachea; background bullous emphysema.
- **Site:** posterior fossa, left cerebellar hemisphere; single lesion shown.
- **T1+C:** large cystic/necrotic lesion with enhancing wall.
- **T2/FLAIR:** fluid-signal component with surrounding oedema.
- **DWI:** heterogeneous; restriction at the rim.

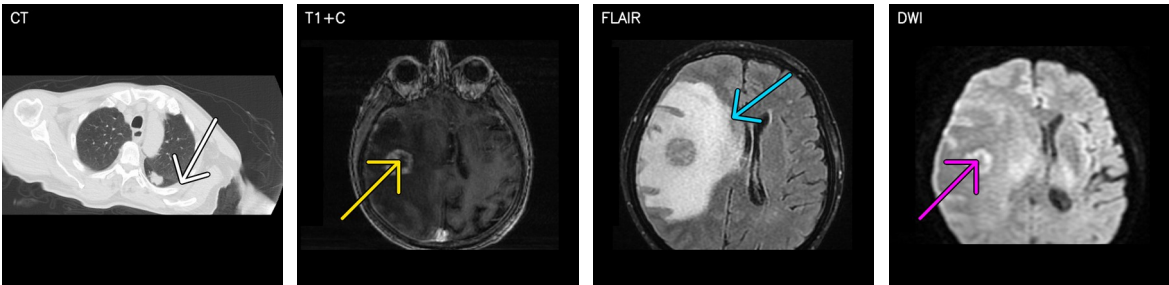
Teaching points

- Squamous BM are less frequent than adenocarcinoma [10]; necrotic or cystic ring-enhancing morphology is described.
- Posterior fossa lesions: assess mass effect and hydrocephalus.
- Single illustrative case (n = 1).

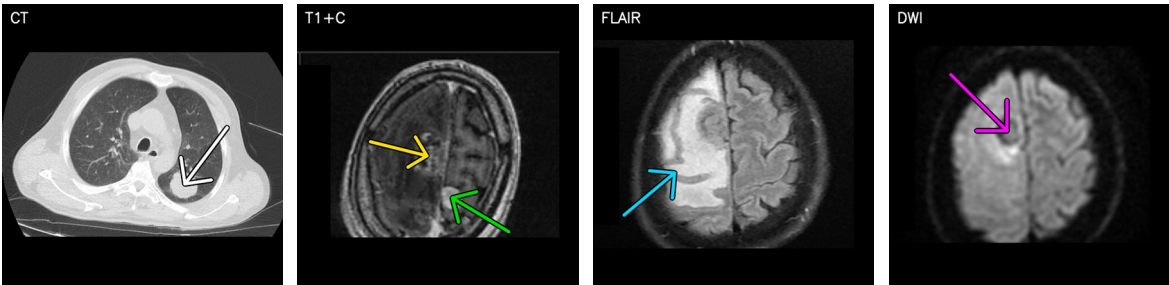
Arrows

- | | |
|------------------|---------------------|
| Enhancing lesion | Vasogenic oedema |
| DWI restriction | Primary lung tumour |

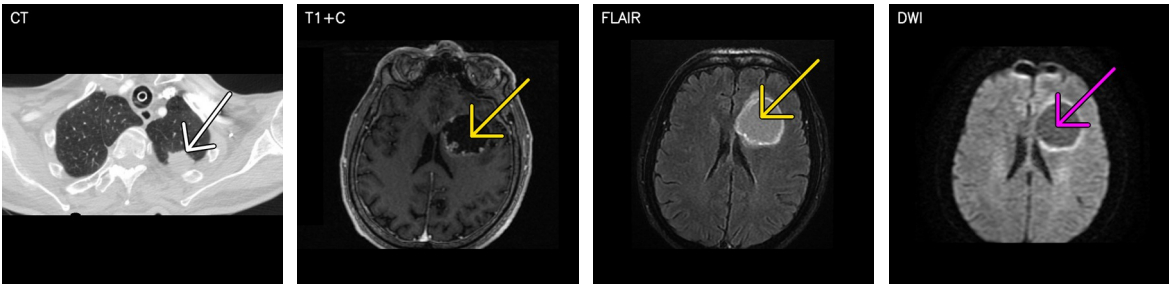
Neuroimaging Spectrum: SCLC & Neuroendocrine Brain Metastases



SCLC-2: chest CT, T1+C, FLAIR, DWI



SCLC-1: chest CT, T1+C (metastasis + incidental meningioma), FLAIR, DWI



LCNEC-1: chest CT, T1+C, FLAIR, DWI

Key imaging features

Feature	SCLC-2	SCLC-1	LCNEC-1
Site	Right frontoparietal	Right paramedian frontoparietal	Left frontal
T1+C	Small ring	Small, heterogeneous	Large necrotic, thick wall
Oedema	Extensive	Extensive	Minimal
DWI	Rim restriction	Focal restriction	Rim restriction

Teaching points

- MRI detects more and smaller lesions than CT: always scan the post-contrast series [5].
- Pitfall (SCLC-1): an extra-axial enhancing lesion may be an incidental meningioma, not a metastasis.
- LCNEC behaves like SCLC; imaging overlaps, so histology cannot be inferred from MRI.

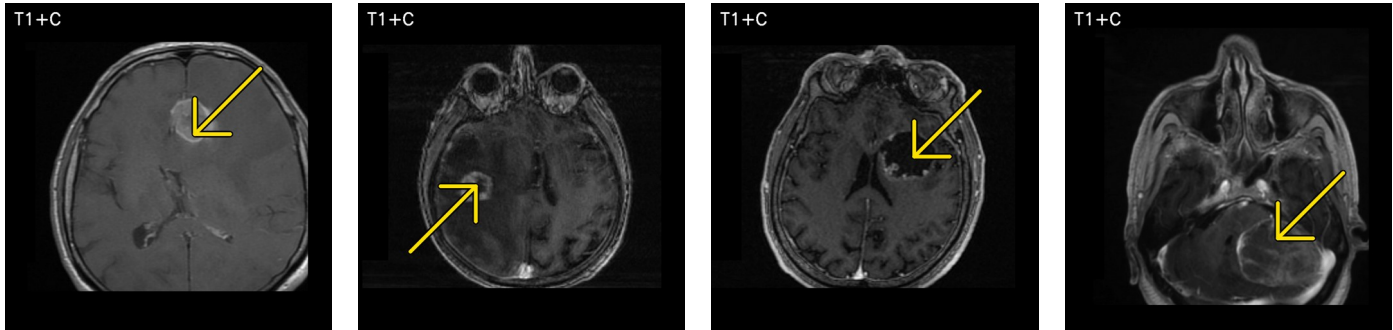
Arrows

- Enhancing lesion
- Vasogenic oedema
- DWI restriction
- Primary lung tumour
- Meningioma

Comparative Matrix

Feature	Adenocarcinoma (2 of 5 shown)	SCLC (2 of 3 shown)	LCNEC (n = 1)	SCC (n = 1)	Literature-expected pattern
Number	1 shown	1 shown	1 shown	1 shown	SCLC: often multiple, 1–3 cm [11,12]; adenocarcinoma: variable
Location	Right temporal; left medial frontal	Right frontoparietal (2/2)	Left frontal	Left cerebellar (posterior fossa)	Grey–white junction, cerebellum, watershed [7,12]
Enhancement	Ring/heterogeneous	Ring or nodular	Cystic/necrotic, thick wall	Cystic, ring	NSCLC necrotic/ring [12]; SCLC homogeneous [11]
Peritumoral oedema	Extensive (2/2)	Extensive (2/2)	Minimal	Moderate	NSCLC: peripheral oedema common [12]
DWI restriction	1/2 (rim)	2/2	Rim	Rim	Variable; frequent in SCLC [11], about 55% of NSCLC [12]; not histology-specific [9]

Substantial overlap between subtypes: no single feature is pathognomonic. One lesion illustrated per case.



T1+C montage: adenocarcinoma • SCLC • LCNEC • squamous cell carcinoma

Discussion



- Biology shapes the imaging phenotype, but overlap is large.

Adenocarcinoma

- Highest BM propensity among NSCLC; frequent in oncogene-driven disease [8]. Marked vasogenic oedema plausibly reflects VEGF-mediated permeability.

SCLC/LCNEC

- High proliferative index and early haematogenous spread [4]; SCLC often shows multiple, small lesions [11]. Contrast MRI is superior to CT for detection (staging, PCI) [5].

Squamous cell carcinoma

- Less frequent BM [10]; necrotic/cystic ring-enhancing lesions described.

This series

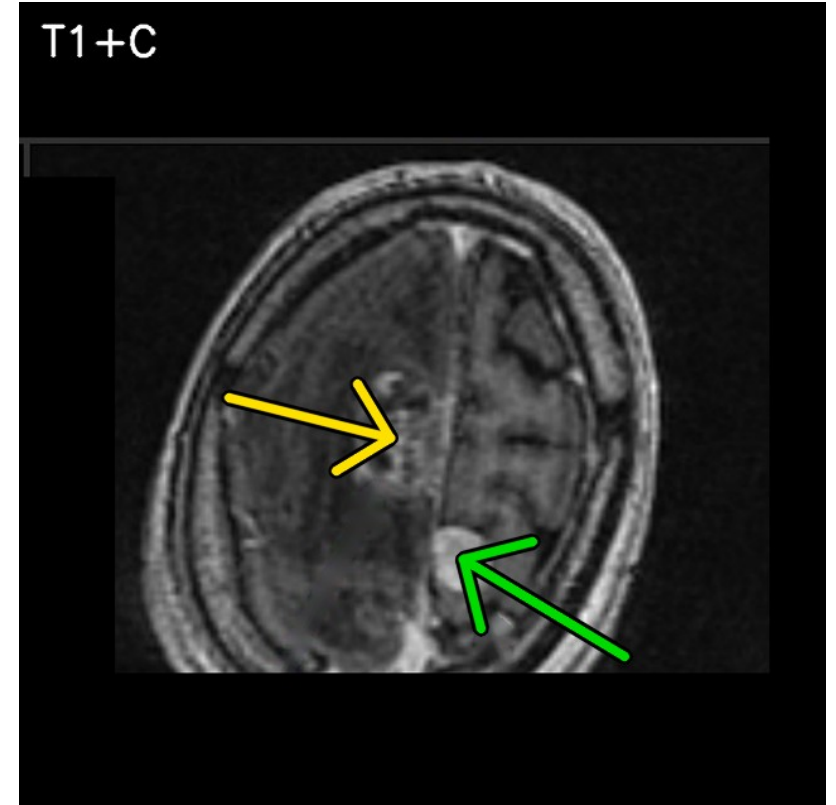
- In the illustrated cases, SCLC primaries were peripheral, not central, with extensive brain oedema; LCNEC was cystic with little oedema. DWI restriction was frequent (5 of 6) but does not predict histology [9].

Pitfalls

- Meningioma, abscess, glioma, radionecrosis, second primary.

Limitations

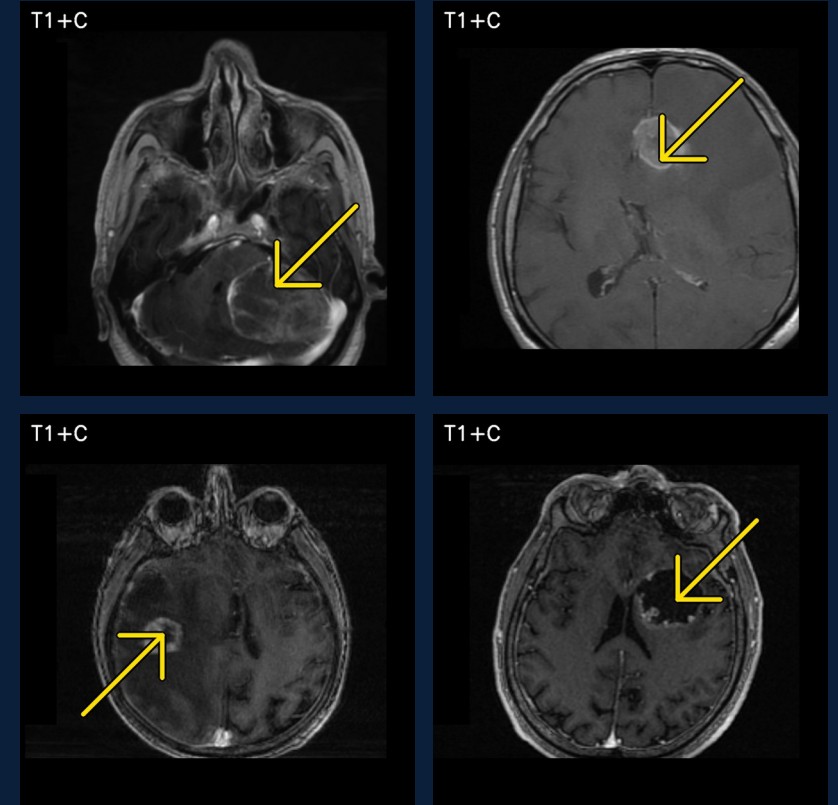
- n = 10 (6 illustrated), retrospective, heterogeneous protocols, no inferential statistics.



Pitfall: metastasis (yellow) vs incidental meningioma (green), SCLC-1

Conclusion & Clinical Take-Home Message

- 1 Contrast-enhanced multi-sequence MRI for lung cancer patients at staging and with new neurological symptoms [6].
- 2 Report lesion number, location, oedema and diffusion: they guide SRS, WBRT and surgery [3,6].
- 3 Adenocarcinoma: look for conspicuous oedema. SCLC/neuroendocrine: look for many small lesions.
- 4 No single feature is histology-specific; beware mimics (e.g. meningioma) and correlate with pathology.



Histology suggests, but does not predict, the MRI pattern.

References & Declarations

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Conflict of interest

The authors declare no conflict of interest.

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Ethics & consent

Ethics committee approval; written informed consent for image publication obtained from all patients.

Image origin

Authors' own clinical material, anonymised; arrows and labels added by the authors.

Contact

[name, email]